



# Integrating Financial Metrics and News Sentiment for Stock Price Forecasting: A CRISP-DM Framework

by

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## Declaration

I, Abhi Tyagi, declared that this Applied Research Project that I have submitted to Dublin Business School for the award of [*MSc Business Analytics*] is the result of my own investigations, except where otherwise stated, where it is clearly acknowledged by references. Furthermore, this work has not been submitted for any other degree.

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## Abstract

Using the CRISP-DM paradigm, this thesis investigates how financial measures and news sentiment might be integrated to improve stock price predictions. Stock markets are notoriously difficult for traditional financial forecasting models to account for because of their non-linearity and extreme volatility, especially when dealing with real-time events and unstructured textual input. In order to overcome methodological constraints and enhance forecast accuracy, this work intends to merge structured financial indicators with sentiment analysis extracted from financial news headlines. The study uses a combination of deep learning (LSTM & CNN) and machine learning (Gradient Boosting) on datasets that include sentiment scores from Reddit news articles and historical stock prices from Yahoo Finance. When combined with technical indications, VADER and TextBlob to measure textual sentiment, allowing for thorough feature engineering. Findings reveal that hybrid deep learning models like CNN-LSTM outperform Gradient Boosting models when it comes to specific equities, such as Microsoft and Apple, while the S&P 500 is a benchmark index where Gradient Boosting models perform better. Validating the benefits of integrating linguistic data, SHAP-based explainable AI algorithms show that sentiment-related characteristics greatly add to prediction accuracy. An approach to stock price forecasting that is both scalable and reproducible is shown here, laying the groundwork for further research into the potential integration of domain expertise and analytics powered by artificial intelligence.

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## CHAPTER 1 – INTRODUCTION

### 1.1 Background

A company sells its fractional ownership via stocks or equity, and the stock market is where the buying and selling of stocks happen(*Stock Market* [no date]). The stock market helps the company raise investment to expand the business and helps investors share the profit these companies earn. Investors are always keen to find companies that generate huge profits, and different financial models are used to study the stock valuation earnings of companies to make any financial decision. As Stock markets are most volatile (Mai 2023), predicting the future depends on factors like pending news or product releases from the company, economic indicators, consumer behaviour, market sentiment, etc. Predicting a stock's price or making any decision is a challenging task.(*Stock Analysis: Forecasting Revenue and Growth* 2025). Analysts rely on fundamental analysis (financial statements, industry trends, economic indicators) and technical analysis (past trading, price movement, and volume) to predict the future movement in stock but lack the integration of real-time factors like news sentiment, global events, social media trends, etc.(Fundamental vs. Technical Analysis: What is the Difference? 2024). These analyses will not add much value to today's dynamic market, and the study suggests integrating Artificial Intelligence (AI) for accurate stock price prediction.

### 1.2 Business Problem

Stock price forecasting is a vibrant research topic that helps people generate huge profits. However, today's high-frequency stock market environment is complex as multiple factors influence the prices(Strauß 2019). Analyzing all the factors is time-consuming and may generate inconsistent results. However, the fundamental analysis may remain the cornerstone of the prediction, but the following fails to add value from market sentiment or any news sources. With massive datasets, company records, and other information available, the study suggests incorporating Artificial Intelligence(AI) to extract actionable insights from various data sources. The advancement in machine learning and deep learning models will help refine these models(Li and Pan, 2022). The pandemic hit the stock market hard, and the whole stock market collapsed. Analyzing such volatility is not in the hands of traditional statistical models. Hence, by understanding the data from various news and social media sources, one can forecast the movement more accurately. Sentiment Analysis, a natural language processing (NLP) based technique, can categorize sentiment as positive, negative, or neutral and help predict stock movement and market trends(*The power of news sentiment in modern financial analysis* 2024). Also, the given technique helps analyze the risk in the early stages. The given integration outperforms various traditional approaches and can help the analyst with real-time sentiment analysis to design strategies.

A 2023 study by (Liapis et al. 2023a) used 67 different features along with stock price data and sentiment data. It tested 30 state-of-the-art algorithms where TCN ( Temporal Convolution Network) outperformed all the other models, showcasing excellent predictive capabilities. The study suggested that adding these features increases the efficiency of models for short-term prediction, and various other models, such as LSTM, saw an increase in accuracy by 12-18%.

Another research by (Arratia et al. 2021) guides how to use sentiment data for investment and trading, as data is the central part of building these models. The study showed that the longer the news is, the more volatility there is. Negative sentiments from news affect the stock price more than positive ones, and the effect is more on mid- and small-size company stocks.(Khonde et al. 2024) showcased the improvement in the accuracy of models like LSTM and in categorizing the sentiment with a mean absolute error of 3.4%.

### 1.3 Research Questions

1. Does combining news datasets lead to superior stock market prediction?
2. How do different deep-learning and machine-learning architectures compare effectiveness in predicting the prices when combined with text data?
3. What is the impact of News data impact solely on stock market prediction?

### 1.4 Research Gap

Based on the given research questions and the problem, our Research falls into the methodological gap(Miles 2017). Traditional methods are insufficient in predicting stock prices, evident by the COVID-19 crisis(Yuan et al., 2023). Hence, integrating financial metrics with the data helps get the correct predictions. With technological advancement, Natural Language Processing (NLP) can be used to understand sentiment and news data, and machine learning models can use data from multiple sources to forecast values.(Antweiler and Frank 2004) It is shown that internet messages are related to the stock market and that their magnitude is high compared to other features.

However, a proper domain understanding is required to build such a model, such as understanding financial metrics, the text data, and how to solve this problem via data mining. Our study will incorporate the CRISP-DM model, which is mainly used in many projects, especially in financial markets(Plotnikova et al., 2022). After all the advancements in data mining capabilities, a research gap remains to standardize combining financial metrics with the news sentiment to optimize the given problem and integrate financial metrics for short- and long-term predictions.

### 1.4 Research aim

The Research aims to add value to the stock price forecasting by integrating the financial news data using sentiment analysis. The study seeks to address the methodological gap by using Natural Language Processing (NLP) and various machine learning and deep learning models to find the impact of news sentiment on stock price data. Implementing the CRISP-DM model study focuses on standardizing the Approach for stock market prediction using linguistic data.

### 1.5 Objectives

- 1 Collect the live data using APIs and integrate them to build the AI-based models per the business problem.



- 2 Compare the performance of machine learning, and deep learning models on financial stock datasets with news sentiment data.
- 3 Assess the effectiveness of deep learning models like Convolutional Neural Network (CNN), Long-Short Term Memory(LSTM) when combined with news sentiment datasets.
- 4 Determine the best Natural Language Processing (NLP) practices to integrate financial and news data and assess its contribution.
- 5 Implement a state-of-the-art Cross Industry Standard Process for Data Mining (CRISP-DM) model to build a structured and reproducible code to improve the reliability of stock price prediction.
- 6 Identify the challenges in integrating real-time financial and textual data for stock price prediction.

## 1.6 Chapter Overview

### Chapter I: Introduction

The chapter shows the problem of stock price forecast prediction and how AI can solve it by integrating various data sources, i.e., Financial News Data.

### Chapter II: Literature Survey

The second chapter shows all the past work done in the Research and improvements in the field of stock price prediction to finally integrate it with the Financial dataset and build the roadmap for the Research and development.

### Chapter III: Research Methodology,

The third chapter showcases the detailed implementation of the CRISP-DM model. It highlights integrating the financial dataset with the stock price dataset using machine learning and deep learning models.

### Chapter IV: Results and Discussion

The fourth chapter shows the results obtained from various metrics and evaluates accuracy metrics to find the benchmark solution.

### Chapter V: Conclusion and Future Work

The chapter summarizes the whole Research, answering all the research questions and setting the roadmap for future discoveries and further enhancement in Research.

## CHAPTER 2– LITERATURE SURVEY

### 2.1 Introduction

Many factors influence stock exchange market management in a complex and dynamic environment. Financial forecasting is a critical component in the stock market that helps investors to buy, sell or hold any stock. These markets are the backbone of any economy. Stock markets are volatile and hit by two of the highest volatile sessions, i.e. in 2008 and then during the pandemic. Both crises tested the prediction power of various traditional methods and showcased the adaptation of advanced technologies like machine learning. Technical and fundamental analysis are limited to handling large datasets and fail to capture non-linear behavior.

On the other hand, predictive analytics excel in analyzing past data, uncovering patterns, and forecasting future trends (Sravani et al., 2024). Fundamental analysis and technical analysis are also critical in predicting stock prices, where fundamental analysis focuses on economic factors and financial results and technical analysis focuses on historical prices and volume of the data (Amini et al. 2015). Research showcases that the models using the fundamental analysis generally outperform the ones using the technical analysis but combining both approaches yields higher results (Beyaz et al. 2018).

With technological advancement and big data, relying on such techniques is not fruitful, and AI implementation has demonstrated higher accuracy than these models. (Chandrashekar et al. 2023) Showcased how a statistical model ARIMA combined with a gradient boosting algorithm attained an accuracy of 96.4% on the S&P 500 index. The ARIMA model was used to pre-process the time series data, and the gradient model then used the residuals and pre-processed data. Another study by (Liapis et al. 2023b) highlighted that incorporating the relevant sentiment analysis data also helps the model to increase its accuracy by introducing better feature space.

### 2.2 Financial Metrics for Stock Price Prediction

To understand the dynamics of the stocks, fundamental and technical analyses are used that help make an informed decision on whether to buy a stock. Fundamental analysis is all about understanding the value of the company using metrics like Price-to-earnings (P/E) ratio, Earning Per Share (EPS), Return on Equity (ROE) etc. (Masyita and Sekartadjie 2016) and technical analysis is used to understand the movement of share using Moving Averages (MA), Relative Strength Index (RSI), etc. in market and making money by timely entry and exit (Rongrodejarnarug 2024). Both of these techniques are combined to attain better results and are suitable based on an investor's goal, whether it is short-term or long-term investment. (Petrusheva and Jordanoski 2016) Showcased the use of technical and fundamental analysis combined for better results. Both of these techniques are pretty opposite to one another. However, investors with fundamental analysis use technical analysis to find the right time to buy or sell a stock. On the other hand, the same strategy is used to understand the stock movement and stock prices using technical analysis at a particular time, and an investment can

be made based on the fundamental analysis.(Petrusheva and Jordanoski 2016). Both analyses may vary in results but are not dominant on top of one another. Hence, using both of them combined can solve the problem. Machine learning models have leveraged these financial metrics to predict the stock price. The fundamental parameters cannot capture most of the information, and studies have shown that deep learning algorithms like LSTM have shown excellent results in predicting the stock price using historical data and various indicators. Also, these models' effectiveness depends on the data fed and the forecasting objectives(Li and Bastos 2020). Multiple data sources can be integrated for stock price forecasting.(Phan and Chang [no date])used a dataset with 269 records from various sectors between 2019 and 2023. The study used the Discounted Cash Flow (DCF) and financial ratios to predict the annual stock price difference and the difference between current stock price and intrinsic value, where the Linear Regression model outperformed models like CNN and LSTM—presented the use of machine learning models on the Belgrade Stock Exchange (BELEX) dataset. The dataset contains many indicators, and the models achieved an RMSE of 0.043%. This accuracy is subject to the low data availability, and increasing the number of records is better to understand the data better. (Peng 2019) integrated oversized data frames like Apache Hadoop to process the sizeable financial dataset. The study used 13 Stocks from the US Oil Fund from Yahoo Finance and forecasted their value.

## 2.3 Sentiment Analysis in Financial Markets

Sentiment Analysis is extracting users' feelings and emotions using Natural Language Processing (NLP). It is analyzed at three levels in the corpus, where the first level is document level analysis where the whole document is analyzed for a positive or negative sentiment. Second is sentence-level sentiment, where the whole sentence is analyzed for sentiment, showcasing the subjectivity of the data. Third is the feature level where the analysis was done on the overall opinion(Joshi and Itkat 2014). Based on this analysis further different sentiment techniques are used. Different machine learning algorithms like Support Vector Machine (SVM), N-gram, Naïve Bayes, KNN and Rule-Based approaches like the based Approach can be used(Devika et al. 2016). However, traditional machine learning algorithms have specific setbacks. Deep learning models outperform machine learning models in sentiment analysis tasks (Zhang et al. 2024). The study used the Amazon customer review dataset and compared machine learning and deep learning algorithms. The RNN model attained an accuracy of 98% with balance precision and recall scores for all the classes. These studies provide strengths and weaknesses of these models as per the business domain and would help provide insights. As these models can understand the text data and classify sentiment in a better way. Understanding the impact of sentiment on the stock price will help combine the NLP with forecast models for better predictions.(Allen et al. 2015) studied the impact of financial news on the stock price data using the Dow Jones Industrial Average (DJIA) and the sentiment score. The study used a multi-factor model and suggested that the sentiment score impacts the company returns. The study used the OLS ( Ordinary Least Square) and QR (Quantile Regression) methods, where the coefficient showcased the significant impact on stock returns.(Cioroianu et al. 2024) Demonstrated the effect of social media sentiment on stock market prices, with excitement sentiment showcasing the highest impact. The study used the VADER-Sentiment Analysis

package and extracted the Vader component. The Information Share (IS) component and other interactions showcased the sedimentation of Kodak's share price change based on the sentiment.(Mujtaba Mian and Sankaraguruswamy 2012) Their study showcased that the market-wide sentiment of investors affects stock prices. Using the Baker-Wurgler Investor Sentiment Index between the period of 1972-2007, the study showcased a higher sensitivity to good news leading to better earnings and vice-versa. This effect was seen on the small, young, volatile, non-dividend-paying stocks. Study the sentiment: various sentiment analysis tools in the market and research have focused on understanding the impact of various techniques in financial prediction models. VADER ( Valence Aware Dictionary and Sentiment Reasoner) and FinBERT have showcased a high accuracy when combined with historical data with the stock price dataset.(He et al. 2024) used the Yahoo Finance dataset and Yahoo Stock Prediction news dataset from Kaggle; using these two technologies, the results show that FinBERT text analysis is better than VADER; however, both showcased better stock price prediction and have their advantages in real life.

## 2.4 Machine Learning and Deep Learning in Stock Forecasting

AI is all about learning from past data, and it excels in capturing complex patterns from the data and forecasting the future. Various machine learning and deep learning algorithms are built based on given properties with their use cases in respective fields.(Hoang Vuong et al. 2024) Random Forest, KNN, SVM, and Bayesian Networks perform more than the basic ARIMA model. ARIMA models are basic statistical models that are good for short and medium-term prediction, but machine learning models ace in prediction accuracy. Also, deep learning models like CNN, RNN and LSTM are extensively used for the same purpose because they can capture sequential information.(Ajiga et al. 2024) their paper showcased the importance of feature selection in predicting the stock market price. Finding relevant features and reducing the noise will increase the predictive power. Techniques like Independent Component Analysis (ICA) and Canonical Relation Analysis (CRA) can be used to improve the accuracy of these models(Guo et al., 2014).

Deep learning models showcase a better performance than the machine learning model as they can capture most of the information from the data with less need for feature selection. The following techniques have shown significant effectiveness in time series and textual data analysis.(Makinde 2024) trained the LSTM model and GRU model with different optimization techniques as compared to the basic training done using Gradient Descent on Yahoo Financial Dataset and showcased that the GRU model with Adam Optimizer showcased the highest accuracy compared to Nesterov Accelerated Gradient (NAG). The following model with Adam optimizer can be used for studies in the real world. Given the practical usage of these models in the real world, their application differs in every aspect.(Jin et al., 2020) showcased the accurate stock price prediction with less time delay using the LSTM models. The high volatility and noise in stock market data in real-world scenarios make it difficult to predict the price. Using the sentiment analysis and emotional tendency of the investor study, Empirical Modal Decomposition (EMD) for better accuracy. The study used a dataset from the Stocktwits

website to extract the sentiment and Apple stock price and using the attention mechanism, trained the LSTM model and received the least errors on all the metrics.

Hybrid models have also shown state-of-the-art results in understanding the market dynamics and improving the accuracy of stock price forecasting. Combining the power of NLP and other sentiment analysis approaches like Lexicon Analysis and advanced sentiment Analysis techniques like LSTM, GRU and RNN and taking information from various data sources such as Twitter, Stock Twits and Reddit, these models tend to perform better(Zheng et al. 2024). More advanced hybrid models and deep learning techniques have also shown better results on different datasets than single models.(Dang et al. 2021) studied different algorithms on many publicly available datasets which are further cleaned and vectorized using TF-IDF, BERT and Word2Vec. These embeddings are tested with the combination of CNN, LSTM and SVM models and combining these models yields a better result than base models, especially with SVM. Combined with hybrid techniques, these hybrid models help reduce the noise in the data and the grammatical errors.(Ahmad et al. 2017) studies techniques, such as pSenti, a concept-level sentiment approach that combines learning-based sentiment classification methods like SAIL, NILC\_USP techniques are discussed, and the study highlights how these techniques, combined with the hybrid approaches, lead to better results. As user-generated content always has some false information and noise.

## 2.5 CRISP-DM Framework in Financial Applications

Cross Industry Standard Process For Data Mining (CRISP-DM) is a widely accepted methodology for data mining projects(Nava and Hernández 2012). It was developed as a standard tool for data scientists and can be used in various big data environments where one wants to extract patterns in the data. (Abasova et al. 2021) It contains 6 phases: business understanding, data understanding, data preparation, modeling, evaluation, and deployment and is an iterative process and can be used in any data mining project. The methodology also follows a hierarchical structure that helps in managing the project and reduces the complexity of the project. CRISP-DM is a widely used methodology that is not domain-specific and can be aligned with any domain. It combines various disciplines, from software engineering to computer science and business intelligence, to deliver industry-standard products. Many studies have used the CRISP-DM model in financial modeling and predictive analytic projects.(Maciej Serda et al. 2013) Used CRISP-DM model to build a classification model that helps detect fraudulent transactions in financial systems. A logistic regression model and the structured approach from the given data mining technique were built. With 200,00 rows and 200 numerical columns, the model achieved an accuracy of 91.46%. Another case study by(Plotnikova et al. [no date]) showcased the use of the methodology in the financial domain and the gap in various steps for flexibility and scalability as per the business problem. The research studied all 6 phases and discussed the gap in evaluation and deployment. In evaluation, one can have further modifications as per business, and deployment may vary from business to business, leading to new strategies to fit in organization contexts. The CRISP-DM model is an ideal methodology for using unstructured sentiment data with the structured data of stock prices. It is an excellent example of data blending, collection, modelling and evaluation. The

methodology will help understand the data and extract the patterns and features from these different data formats, addressing issues like data quality, missing data and any other inconsistency. The methodology provides a structured way to solve the given problem, from finding the business context to finding ways to join both datasets and prepare them for modelling. The data preparation steps would differ for both types and follow different paradigms like NLU ( Natural Language Understanding) and EDA (Exploratory Data Analysis) for unstructured and structured data. Ultimately, the models would be built and evaluated on the constant metrics, and a final output and novelty would be given for the research.

## 2.6 Research Gaps and Motivation

Integrating financial metrics and stock price data with sentiment data is highly researched. However, integrating the CRISP-DM model for forecasting the stock price is an emerging area of research. The past research showcases the advantages of technical and fundamental analysis and uses machine learning and deep learning models to predict stock prices. However, using the CRISP-DM model, a data mining methodology for predicting stock prices, is still not standardized. All the research showcases the isolation of these models but fails to add the context of a structured approach for the given modelling technique. The given gap limits the reproducibility and scalability of a given model in price forecasting. Also, there are many challenges in integrating financial and textual data. Textual data, which is highly unstructured and high dimensional, faces data cleaning challenges and integration with the stock market data. Forecasting models fail to maintain the word's meaning and face methodological challenges integrating it with the financial world. Our research uses real-time datasets from Yahoo Finance and Reddit News. It uses domain knowledge to extract and clean the data, leading to better data preparation that will help in better predictions. From traditional models to advanced deep learning models, the research focuses on building these models using the CRISP-DM framework that will address the reproducibility and scalability of the models. Also, the standardized methodology will help integrate the data sources and evaluate them on consistent metrics for better results.

## 2.7 Conclusion

The main conclusion drawn from the study are as follows:

- **Complexity and Volatility of Stock Market:** Modern stock markets are complex and nonlinear, making traditional methods like fundamental and technical analysis inadequate to understand them. This is particularly true during times of extreme volatility, like the financial crisis of 2008 or the COVID-19 pandemic
- **Integrating Financial Metrics and Sentiment Analysis:** Prediction accuracy is much improved when financial indicators (such as P/E ratio, EPS, ROE) are combined with sentiment data obtained from news and social media. Unlike conventional models, sentiment research backed by NLP can reveal the current state of the market in real time.
- **Machine Learning and Deep Learning Capabilities:** The most recent developments in deep learning and machine learning have led to state-of-the-art models like LSTM and GRU outperform their predecessors when it comes to predicting stock values.

This is especially true when these models are fed multi-source data, which includes textual sentiment.

- **Methodological Gap:** Despite these developments, there is still a lack of replicable, defined frameworks for methodologically integrating various data sources and analytical techniques. Hence, forecasting models can't scale well or be trusted.

Stock price prediction is complex and involves many factors; so, a systematic and iterative technique is necessary. Throughout the data mining lifecycle, from gaining a business knowledge and preparing the data to modelling, evaluating, and deploying the results, the strong framework provided by CRISP-DM is invaluable. It improves the reliability of models by establishing a clear and repeatable process for integrating sentiment data with financial measures. In the ever-shifting financial markets, the capacity to quickly adapt to new data sources and modelling methodologies is essential, and this framework provides this ability. Better Market Dynamics Capture and Actionable Insights from CRISP-DM-Driven Models by Systematic Incorporation of Domain Knowledge and Iterative Evaluation, Leading to Better Decisions.

Table 1 Comparative Study

| Title                                                                                                      | Year | Authors              | Method Used                            | Results Achieved                                                                  |
|------------------------------------------------------------------------------------------------------------|------|----------------------|----------------------------------------|-----------------------------------------------------------------------------------|
| Enhanced Investment Strategies Through Predictive Analytics of Stock Prices.                               | 2024 | Sravani et al.       | LSTM & RNN                             | Achieved higher accuracy than base models                                         |
| Machine Learning Prediction Approach for Financial Forecast System in Stock Exchange Marketing Management. | 2023 | Chandrashekar et al. | ARIMA & Gradient Boosting              | 96.4% Combined accuracy achieved                                                  |
| Investigating Deep Stock Market Forecasting with Sentiment Analysis. <i>Entropy</i>                        | 2023 | Liapis et al.        | 30 Different ML and DL models are used | Showcased the prevalence of method and improvement when used with sentiment data. |

|                                                                                                               |      |                                                                                                 |                                                                     |                                                                                                         |
|---------------------------------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
|                                                                                                               |      |                                                                                                 |                                                                     |                                                                                                         |
| Stock Market Forecasting Using Deep Learning and Technical Analysis: A Systematic Review.                     | 2020 | Li, A.W. and Bastos, G.S.                                                                       | Evaluate Deep Learning models using Technical Analysis              | LSTM were used in 73.5% of the studies.                                                                 |
| Comparative Study of Sentiment Analysis on Customer Reviews Using Machine Learning and Deep Learning.         | 2024 | Zhang, Y.; A., Ashbaugh,                                                                        | LR, Naïve Bayes, Random Forest, RNN, CNN                            | RNN gave a balanced accuracy of 98%                                                                     |
| Daily Market News Sentiment and Stock Prices.                                                                 | 2015 | Allen, D.E., Mcaleer, M. and Singh, A.K.                                                        | OLS ( Ordinary Least Square) and QR (Quantile Regression)           | Showcased the impact of sentiment on financial dataset                                                  |
| Exploring emotional sentiment to understand the market response to unexpected corporate pivots.               | 2024 | Cioroianu, I., Corbet, S., Hou, Y. (Greg), Hu, Y., Larkin, C. and Taffler, R.                   | IS, ILS and CS Information share measures                           | Showcased the effect of excitement on stock prices.                                                     |
| Sentiment to Stocks: Rule-Based and Deep Learning Sentiment Analysis for LSTM-Driven Stock Prediction.        | 2024 | He, S., Meng, Q., Chen, B., Jiang, X. and Gao, S.                                               | FinBERT and VLADER                                                  | Showcased a high correlation between sentiment and stock prices based on sentiment analysis techniques. |
| A bibliometric literature review of stock price forecasting: From statistical model to deep learning approach | 2024 | Hoang Vuong, P., Hung Phu, L., Hong Van Nguyen, T., Nhat Duy, L., The Bao, P. and Dat Trinh, T. | Compared Traditional Models with Machine Learning and Deep Learning | It Showcased that the efficiency of deep learning is better than Machine learning followed by           |



|                                                                                                                                                                |       |                                                                                                              |                                         |                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                |       |                                                                                                              |                                         | Traditional Models.                                                                                                   |
| MACHINE LEARNING FOR STOCK MARKET FORECASTING: A REVIEW OF MODELS AND ACCURACY.                                                                                | 2024. | Ajiga, D.I., Adeleye, R.A., Tubokirifuruar, T.S., Bello, B.G., Ndubuisi, N.L., Asuzu, O.F. and Owolabi, O.R. | NA                                      | Showcased the importance of Feature Selection in improving the accuracy of Machine Learning and Deep Learning Models. |
| OPTIMIZING TIME SERIES FORECASTING: A COMPARATIVE STUDY OF ADAM AND NESTEROV ACCELERATED GRADIENT ON LSTM AND GRU NETWORKS USING STOCK MARKET DATA A PREPRINT. | 2024  | Makinde, A.                                                                                                  | LSTM and GRU                            | GRU outperformed LST with Adam and NAG Optimizer.                                                                     |
| Stock closing price prediction based on sentiment analysis and LSTM.                                                                                           | 2020  | Jin, Z., Yang, Y. and Liu, Y.                                                                                | LSTM with Empirical Modal Decomposition | Attained least error with LSTM model as compared to other models                                                      |
| Hybrid Deep Learning Models for Sentiment Analysis                                                                                                             | 2021  | Dang, C.N., Moreno-García, M.N. and De La Prieta, F.                                                         | CNN, LSTM,SVM                           | Showcased the supremacy of hybrid models as compared to base models.                                                  |

## CHAPTER 3– METHODOLOGY

### 3.1 Introduction

The chapter showcase methodology implemented to investigate and develop a stock price forecasting model by integrating financial metrics with the news data. For the application the study has used a structure metrology known as CRISP-DM. Cross Industry Standard Process for Data Mining is a framework that is widely used in data mining and predictive analytics.(*CRISP-DM* |*Data Science PM* 2024) The study highlights each phase of the CRISP-DM model and showcase how numerical data when combined with unstructured textual data enhances the accuracy of stock predictions. The chapter details data sources, techniques, and algorithms used for the research and highlights the preprocessing steps to integrate textural data with the stock price data.

### 3.2 CRISP-DM

The Cross Industry Standard Process for Data Mining is a structured framework that is used in various industries for structured implementation of data mining and predictive analytics. The following methodology is an iterative process and is a six-phase approach where every phase is interlinked to one another based on their usage(Clinton et al. 1999). This sequence of all the phases is in order and moving back and forth is required based on the type of problem. The model is iterative and cyclical in nature. The reason it is chosen is as it is the most followed approach in most of the data science projects around the world and it provides a structured way to integrate the financial data with the unstructured text data(*The Continued Dominance of CRISP-DM in 2024* |*LinkedIn* 2024).

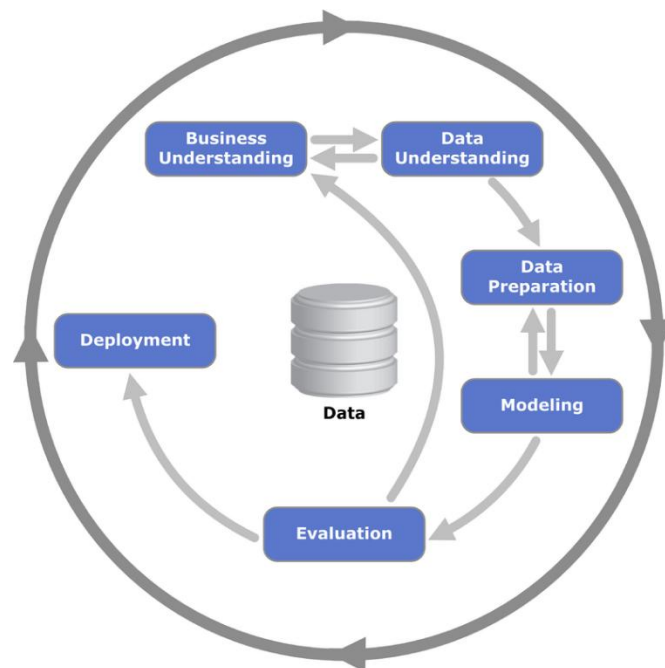


Figure 1 CRISP-DM

There are six phases in the CRISP-DM model, all are described as follows:

**Business Understanding:** The first phase describes the understanding and objective of the business and tries to convert the business problem to a database problem. In our research, the main objective is to improve stock price movement by incorporating the new data.

**Data Understanding:** The next phase involves collecting and exploring the relevant data and gaining insights from the data regarding the data quality. Here, it includes identifying the financial variables, collecting the news data, and conducting analysis to detect patterns(Ayele 2020).

**Data Preparation:** Data preparation is the most important step in any data mining project. This phase covers all the activities to construct the final data set for modeling. It includes cleaning and transforming the news data as well as time series data and combining them for machine learning and deep learning models(Li 2019).

**Modeling:** In this step, the modeling techniques were selected and applied to the prepared data. Here, different predictive models are trained, and their parameters are tuned for optimized performance.

**Evaluation:** This phase assesses the effectiveness of machine learning models using appropriate performance metrics. Here, all the models are compared and validated as per the business objective, and the right model is selected.

**Deployment:** The final phase involves the implementation of a predictive model in the real world or simulated world. Proper documentation is generated to explain the implementation of the predictive model in our research. The outputs are being explained using explainable artificial intelligence (XAI) for better decision-making

## 3.2 Data Source and Collection

The data used in the research is Reddit News dataset having 73608 news articles. This data is text data and the sock price dataset is taken from Yahoo Finance (*yfinance* · *PyPI* [no date]). A special library designed to get the market data in Python to generate real-time insights.

Loading and processing news data...

| News  |                     |                                                                                                                                                                                          |
|-------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Index | Date                |                                                                                                                                                                                          |
| 0     | 2016-07-01 00:00:00 | A 117-year-old woman in Mexico City finally received her birth certificate, and died a few hours later. Trinidad Alvarez Lira had waited years for proof that she had been born in 1898. |
| 1     | 2016-07-01 00:00:00 | IMF chief backs Athens as permanent Olympic host                                                                                                                                         |
| 2     | 2016-07-01 00:00:00 | The president of France says if Brexit won, so can Donald Trump                                                                                                                          |
| 3     | 2016-07-01 00:00:00 | British Man Who Must Give Police 24 Hours' Notice of Sex Threatens Hunger Strike: The man is the subject of a sexual risk order despite having never been convicted of a crime.          |
| 4     | 2016-07-01 00:00:00 | 100+ Nobel laureates urge Greenpeace to stop opposing GMOs                                                                                                                               |

Show 25 per page

Like what you see? Visit the [data table notebook](#) to learn more about interactive tables.

Figure 2 Text Data

Both these datasets are of different formats and require different types of cleaning to be merged, which is discussed in the next chapter.

## 3.3 Data Cleaning

### Text Data

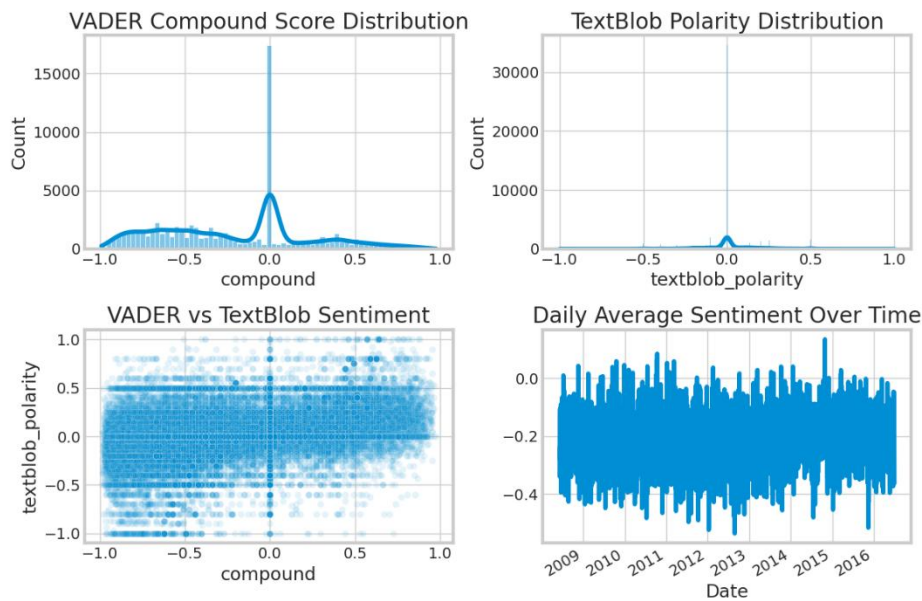
Cleaning text data is way much different from analyzing the normal tabular data. For text data cleaning Natural Language Understanding(NLU) is used and the data is removed from all the discrepancies. To find the better insights from the text data VADER ( Valence Aware Dictionary and Sentiment Reasoner) is used.(*VADER sentiment analysis (with examples)* | Hex [no date]) It is a lexicon and rule-based sentiment analysis tool that detect and quantify the sentiment expressed in text. This method use a dictionary of words and phrases having sentiment score and is applied to the data set for sentiment score based on intensity modifiers such as capitalization, punctuation etc. The model is known for its transparency, speed and effectiveness in short and informal text(*Basic Sentiment Analysis Using R with VADER* || *Medium* 2024). The study has been used to find the compound sentiment and the probability assigned to each sentiment i.e. positive, negative and neutral. The news dataset is then filtered based on the financial keywords, and the data is grouped based on the various sentiment score. There are a total of 4704 financial new headlines out of overall data and the date ranges from 2008-06-08 to 2016-07-01.

### **Stocks Data**

Stocks data is fetched from the Yahoo Finance Library and the data is fetched for S&P500, Microsoft and Apple. The relevant data like Date, Volume, High, Low and Open is used and the moving average for 5 days, 10 days and 20 days is calculated. The Relative Strength Index is also calculated to measure the price momentum. Also, the same data is scaled for better training of machine learning models.

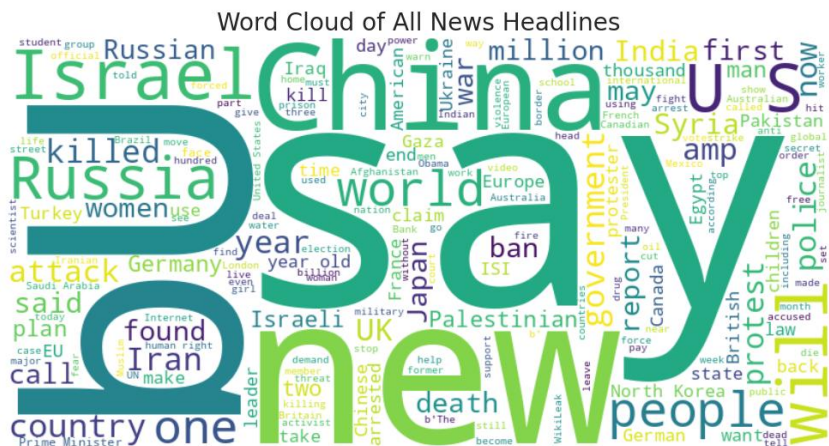
### **3.4 Exploratory Data Analysis**

Once the data is loaded and prepared, the next step is to explore the data and find patterns in it. This can be done by creating summaries of the data or by visualization. As the data is either text or time series. Visualizations are used to understand the data.



The sentiment distribution chart showcase that the most scores are centered towards 0 showcasing that the text data is neutral for Compound Scores as well as Polarity. These charts overall showcase that the sentiment is neutral but when comparing VADER with Textblob there is a correlation between the data but not much. And a visisble decline can be seen between the years 2014-2016(Uhr et al. 2014; Patil et al. 2024).

The next chart showcase word-cloud generated for top 200 words in the news dataset and words like say, China, new has more dominance as compared to other words.



But when looked at the word cloud for Financial News Headline dataset Bank ,China, Economy, government and trade are the most common words showcasing frequent topics in financial news.

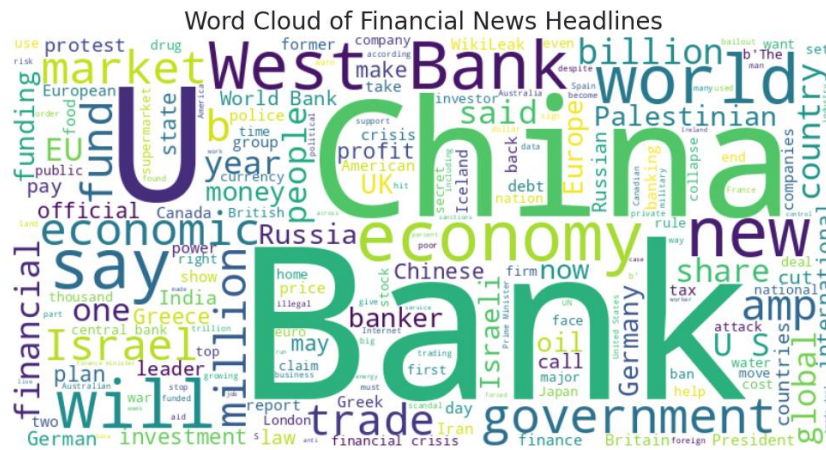


Figure 5 WordCloud For Financial News Data

Next the stock data is analyzed using the visualizations and the patterns observed are as follows:

**S&P 500** stock performance over the year is at an uptrend after the crisis of 2008-2009 the stock showcase consistent uptrend and the volume showcase the same pattern and is higher. However, the new sentiment is almost neutral, having max value of 0.8 on the overall data.



Figure 6 Stock Analysis for S&P 500

The research is also studying all the technical indicators, and one can see that the MCD and RSI are highly correlated with each other. However, the sentiment showcase less correlation with the stocks data.

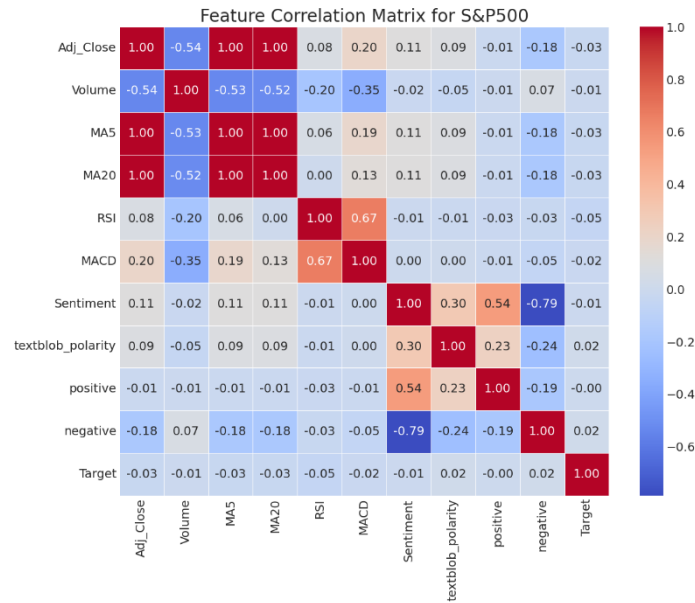


Figure 7 Correlation Matrix S&P500

**Apple** stock prices showcase a steady uptrend and noticeable surges in 2012 and 2014. The moving averages also showcase a constant trend following the actual price. The trading volume is high at the crisis time in 2008-2009 and sentiment is almost neutral with some spikes. However, the stock is more volatile compared to the S&P 500.

Apple Stock Price, Volume and News Sentiment

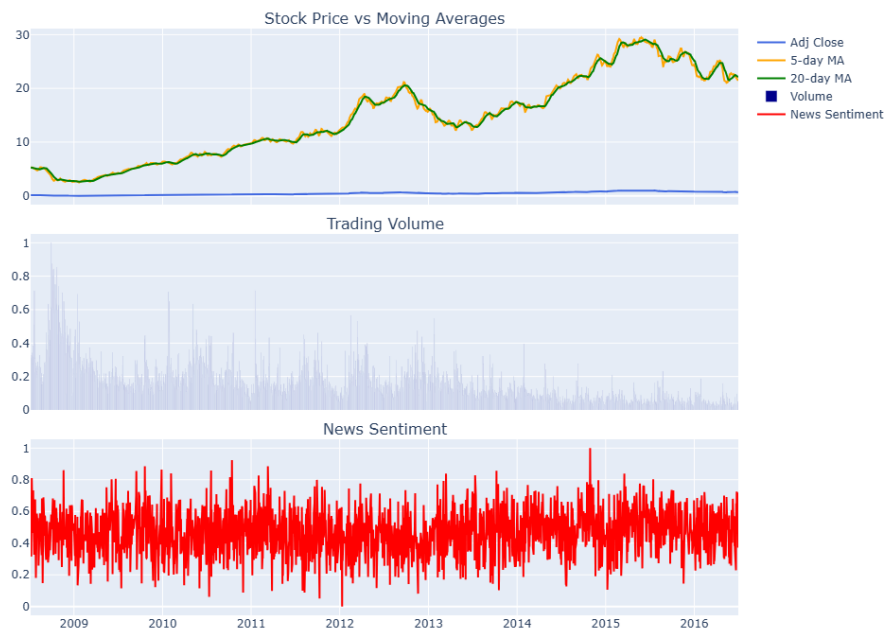


Figure 8 Stock Analysis for Apple

Adj\_close is highly correlated with the moving averages for 5 and 20 days and volume showcased negative correlation with the price and moving averages.



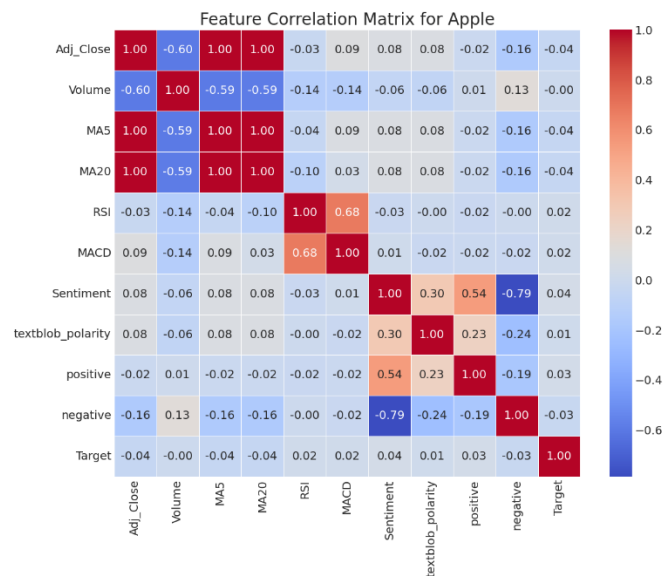


Figure 9 Correlation Matrix Apple

**Microsoft** stock price also showcase an upward trend, and the moving averages showcase stable trends with not much volatility. Similar to the apple, stock shows high spikes in 2009-2014 and an uptrend. But the trading volume saw a decrease over time. The sentiment is also centered to 0.5



Figure 10 Stock Analysis for Microsoft

The heatmap shows a high correlation between the moving averages and ad\_close column. The target variable show weak correlation with the data requiring the need for complex models.



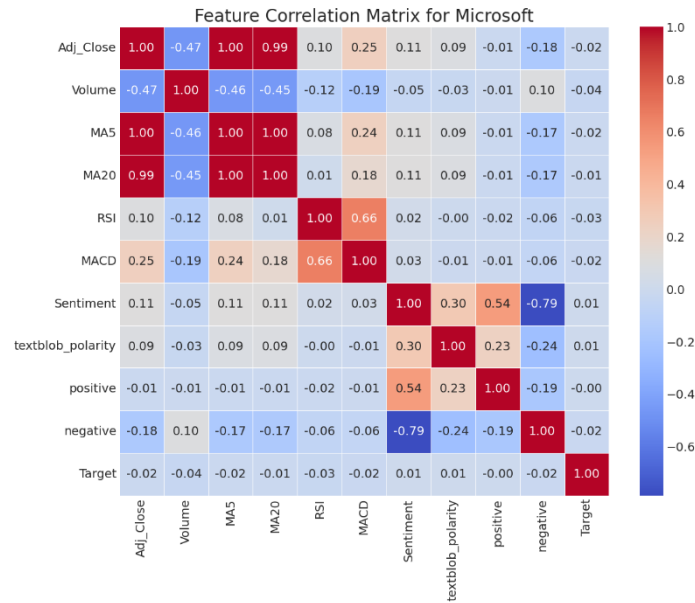


Figure 11 Correlation Matrix Microsoft

### 3.5 Feature Selection and Engineering

The past analysis showed data exploration using visualization to understand the data in a 360-degree approach. To train the models, data needs to be prepared. The time series data is prepared by creating the sequence to be given to the deep learning models using the lookback function, a sliding window would be created to generate input sequences. The three stock datasets are used to generate two different types of stock data, one with basic split and one with extended split(Gamboa 2017; Ghahramani and Najafabadi 2022). The data is further divided into training with 70% data and validation dataset with 15% of the data to train the models.

### 3.6 Model Selection

Different modeling techniques are used in the research to compare the results between machine learning and deep learning models. The models used in the research are defined as follows:

#### GradientBoost

As part of its ensemble learning strategy, the Gradient Boost Classifier (GBC) builds a series of decision trees, with the goal of improving upon each one that came before it(Nakagawa 2018). It creates a more accurate and reliable model by combining the predictions of different trees. In stock price forecasting, GBC shines because it can handle complex, non-linear relationships between inputs (such sentiment and financial indicators) and outputs (like stock prices). This is particularly true when integrating these two types of data. This technique is great at predicting stock movements based on a wide variety of parameters because it efficiently catches complex patterns and relationships within the data. Improved financial forecasting and decision-making are outcomes of GBC's analysis of the association between structured financial data and unstructured news sentiment, which is a common use of the CRISP-DM paradigm in this field(Nakagawa and Yoshida 2022a; Yuvaraj et al. 2025).

## **LSTM**

To model sequential data and capture long-term dependencies, recurrent neural networks (RNNs) like Long Short-Term Memory (LSTM) are trained. The unique architecture of LSTM allows it to remember crucial details for longer periods of time while forgetting unimportant details.(Nadif et al. 2022) This is in contrast to typical RNNs, which lack these features. The architecture of LSTM contains input, forget, and output gates. LSTM's ability to learn from trends and patterns over time makes it ideal for sequence prediction and time series forecasting jobs like stock price forecasting. (Jadoon et al. 2024)One use of LSTM is in stock price prediction, where it can analyze financial parameters and historical stock data to foretell how the price will go in the future. Through the process of learning complicated, time-dependent correlations in the data, it can enhance forecasting accuracy by incorporating other inputs such as news sentiment. The capacity of LSTM to represent these sequential relationships renders it an exceptionally potent instrument for prediction within fields like as healthcare, finance, and natural language processing(Sunday 2024)

## **CNN-LSTM**

When dealing with data that only has one dimension, such a time series or a sequential set of numbers, a convolutional neural network (CNN) is the way to go. 1-D convolutional neural networks (CNNs) use convolutional filters applied over input sequences to identify local patterns or features, as opposed to the more conventional 2-D CNNs utilised in image processing. Stock price predictions, sensor data analysis, and signal processing are examples of jobs that benefit from the temporal or sequential structure of the input data, which these networks excel at.(Patil et al. 2023)As it scans over data and detects essential features, a 1-D CNN can learn patterns from time-series data, such as price trends, volume changes, and other financial variables, which can be used for stock price prediction. Common neural network models (e.g., LSTM) or fully connected layers receive the output and process it further.

The CNN-LSTM model integrates the best features of convolutional neural networks (CNNs) and long short-term memories (LSTMs). The convolutional neural network (CNN) component uses convolutional layers to automatically extract local or spatial properties from sequences or time series as input data.(Hamoudi and Elseifi 2021; Shi et al. 2022) An LSTM network then takes these traits and uses them to identify data dependencies across longer periods of time. The convolutional neural network (CNN) component effectively identifies local patterns (such as trends, peaks, or other short-term characteristics in stock data) and the long-term dependencies and trends over time (LSTM component), making this hybrid architecture an effective tool for tasks such as stock price forecasting. Accurate forecasts in time-dependent, dynamic data, such as stock prices or financial markets, require the model to learn both short-term and long-term patterns. This combination enables the model to do just that.

## **3.7 Model Evaluation and Interpretation**

Models when trained are further evaluated based on various metrics and for the explainability of the model's explainable artificial intelligence (XAI) is used. The metrics and frameworks used are discussed as follows:

## Evaluation Metrics

- **Confusion Matrix:** Confusion matrix is a 2D matrix that compares the actual class labels by predicted class labels in the model. The main terms used in confusion matrix are:
  - True Positive:** Positive class that is correctly predicted.
  - True Negative:** Negative class that is correctly predicted.
  - False Positive:** Positive prediction done for negative class.
  - False Negative:** Negative prediction done for a positive class.
- **Accuracy:** It is the proportion of total correct predictions i.e. (TP+TN) done by the model out of all the predictions.
- **F1-Score:** F1-score is the harmonic mean between the precision and recall where precision is the ratio of actual positives from total positives and recall which is actual positive identified from the model
- **ROC-AUC:** Receiver Operating Characteristics – Area Under the Curve is a chart between the Recall and Specificity that showcase how good the model is behaving in predicting both the classes accurately. With an AUC of 1 indicates perfect classification.

## XAI

Explainable artificial intelligence is the set of methods and techniques that are used to reverse engineering on machine learning models because of their black-box nature. The techniques converts complex black box models into interpretable forms, validating the trust of users and improving the AI systems by understanding how input features influence predictions. The study has used SHAP, also known as the shapely additive explanation. A popular XAI technique, the technique is based on the game theory where each feature is treated as a player contributing to the final prediction of the model. The SHAP calculates feature attributions by considering all the combinations of features and providing the local explanations as well as the model explanations. It can capture non-linear relationships and generate visualization to understand the data.

## CHAPTER 4– RESULTS AND DISCUSSION

### 4.1 Results

The study has utilized various algorithms to train the model where one algorithm is a machine learning model which is considered as baseline model and rest are deep learning models. The results for them are as follows:

#### Gradient Boosting Classifier

Gradient Boosting classifiers are known for their state-of-the-art results in the tabular format. But when applied with the Time Series data they can show effectively good results if the data is windowed(Nakagawa and Yoshida 2022b). Our dataset is windowed at 10 steps and the whole model is trained on it. The model attained an accuracy of 56.1% for the S&P 500. For Apple the results are 46.8% for Accuracy, 52.7% for F1 and 50.9% for ROC was obtained. And the accuracy for Microsoft was 50.2%, with an F1-score of 54.3% and ROC of 51.4%

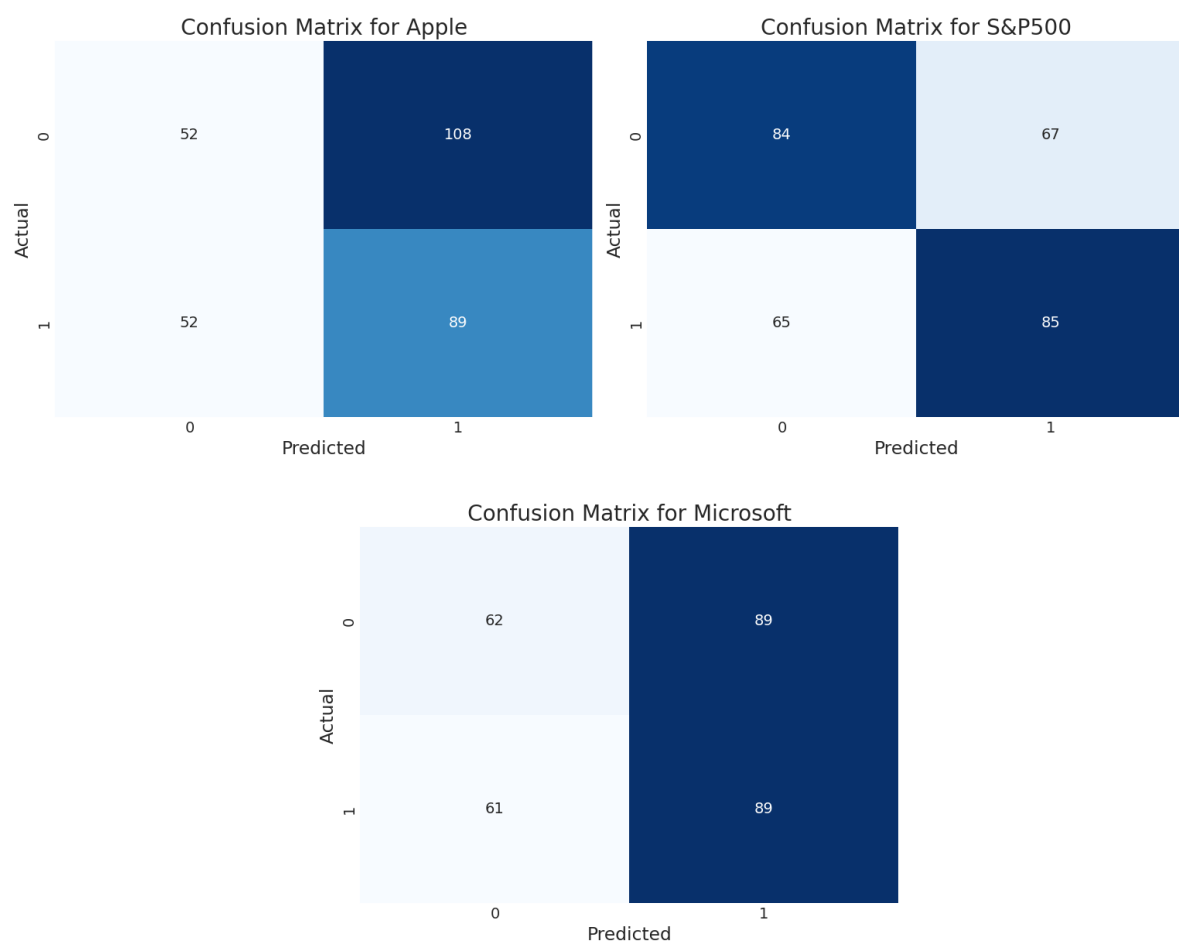


Figure 12 Confusion Matrix Gradient Boost Classifier

Table 2 Accuracy Metrics Gradient Boosting

|           | Accuracy | F1    | ROC   |
|-----------|----------|-------|-------|
| S&P 500   | 56.1%    | 56.3% | 57.1% |
| APPL      | 46.8%    | 52.7% | 50.9% |
| Microsoft | 50.2%    | 54.3% | 51.4% |

## LSTM

The next model that is trained in the research is LSTM model. The model predicts the movement of stock whether it is up or down. The configuration of the LSTM model is as such that the total units of LSTM model are 64 and the dropout is 0.2. This means that at every training step the data removed from the is 20%. The total number of units in each layer are 32. The LSTM model is further configured in different configurations and is trained on the dataset. A huge number of parameters are trained on the model and the results obtained are as follows:

Introduction of the dropout layer helped in consistent results as accuracy is good on such dataset. The baseline model is stable compared to other models but haven't improved much. The deep learning model showed better results but risks overfitting on the data.

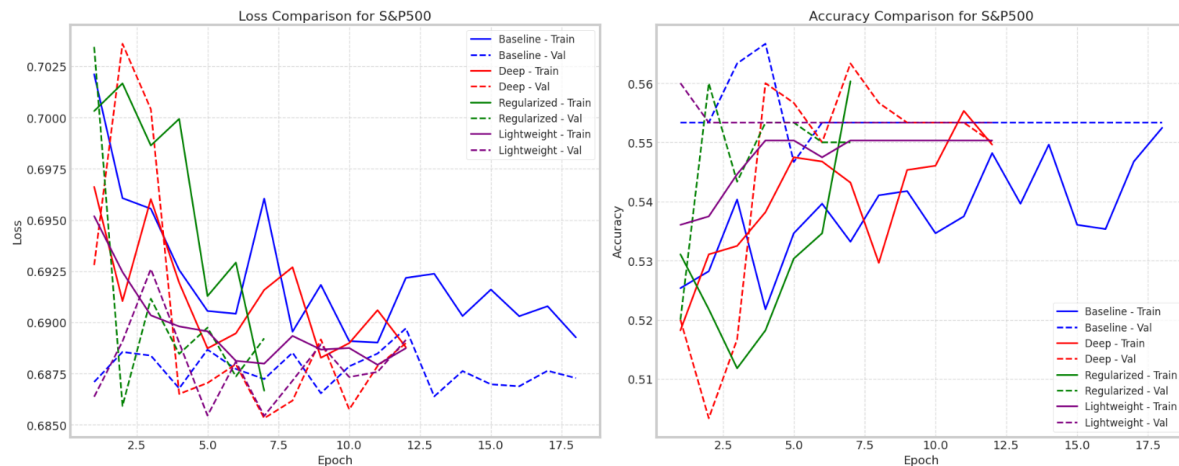


Figure 13 LSTM Analysis -1

The chart for Apple showcase consistent decrease in training loss but the fluctuation in validation loss is minor. The addition of regularized layer help in reducing the overfitting but overall, the lightweight models performed the best.

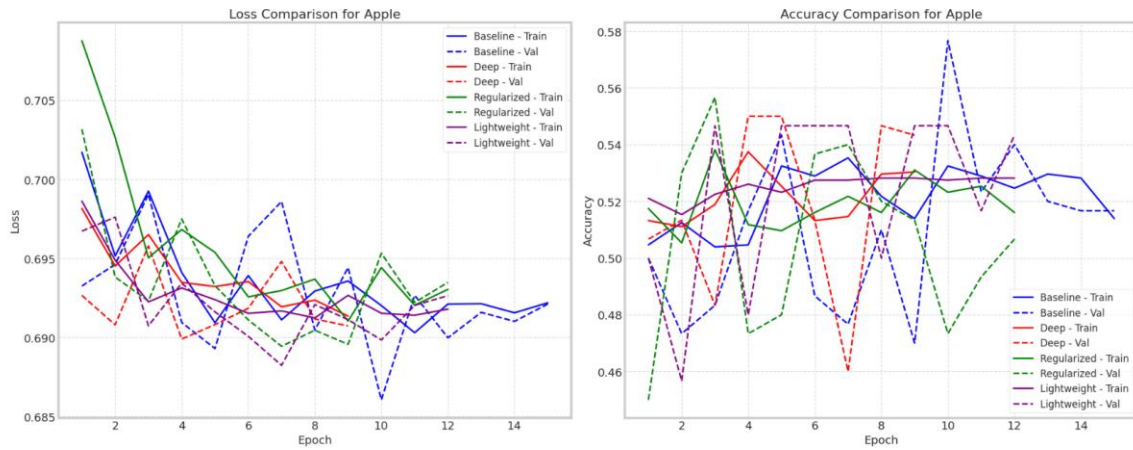


Figure 14 LSTM Analysis -2

Microsoft data when trained with the LSTM model showed a decline in the training loss for baseline model but with higher validation loss indicates overfitting. When looked at on an overall basis all models show a decline in loss showcasing learning is happening. The lightweight model performs well as compared to other models. Deep learning models on the other hand showcased high accuracy but slightly overfitted model.

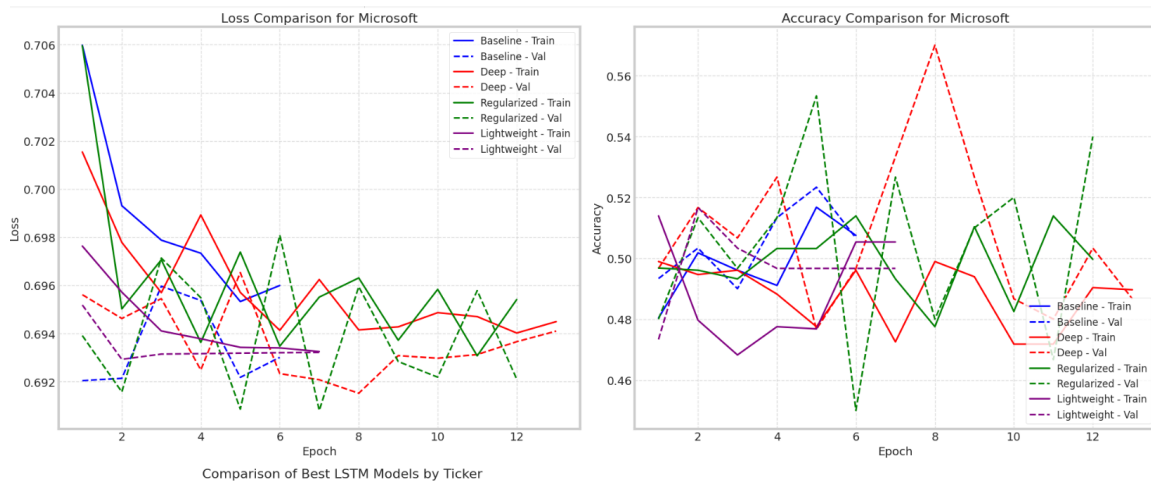


Figure 15 LSTM Analysis -3

## CNN-LSTM

The last model that is trained on the given dataset is CNN-LSTM. CNN that is well known for image data and LSTM for capturing sequential data, are combined together. The CNN-LSTM set up showcase two models trained. One is CNN and the other is LSTM+CNN. The CNN-LSTM model showcased an accuracy of 0.498 and F1 score of 0.665. The accuracy score is slightly lower as compared to S&P 500. Both the models used 64 filters with a dropout of 0.5. Microsoft showcased a bit higher test accuracy as compared to other models

| Ticker    | Model Type | Accuracy | F1 Score | ROC AUC | Filters | Kernel Size | Dropout | Learning Rate | Dense Units | CNN Filters | LSTM Units |
|-----------|------------|----------|----------|---------|---------|-------------|---------|---------------|-------------|-------------|------------|
| S&P500    | CNN        | 0.498    | 0.665    | 0.500   | 64.0    | 3.0         | 0.3     | 0.001         | 32          | nan         | nan        |
| S&P500    | CNN-LSTM   | 0.498    | 0.665    | 0.500   | nan     | nan         | 0.3     | 0.001         | 32          | 128.0       | 32.0       |
| Apple     | CNN        | 0.468    | 0.638    | 0.555   | 64.0    | 3.0         | 0.5     | 0.0005        | 64          | nan         | nan        |
| Apple     | CNN-LSTM   | 0.468    | 0.638    | 0.566   | nan     | nan         | 0.5     | 0.0005        | 64          | 64.0        | 64.0       |
| Microsoft | CNN        | 0.502    | 0.665    | 0.542   | 64.0    | 3.0         | 0.5     | 0.0005        | 64          | nan         | nan        |
| Microsoft | CNN-LSTM   | 0.502    | 0.665    | 0.524   | nan     | nan         | 0.5     | 0.0005        | 64          | 64.0        | 64.0       |

Figure 16 CNN LSTM Results

## 4.2 Evaluation

The evaluation phase of the CRISP-DM model showcase the assessment of all the algorithms developed on the basis of evaluation metrics as well as the business objectives.

- For S&P500 the Gradient Boosting algorithm attained the highest accuracy and rest all other model showcased consistent results in terms of accuracy.
- The F1-score for all the other models except Gradient Boosting was good showcasing that deep learning models provide better prediction in imbalanced dataset.
- With the highest AUC\_ROC score and accuracy score, the Gradient Boosting Algorithm is the best model for S&P 500.

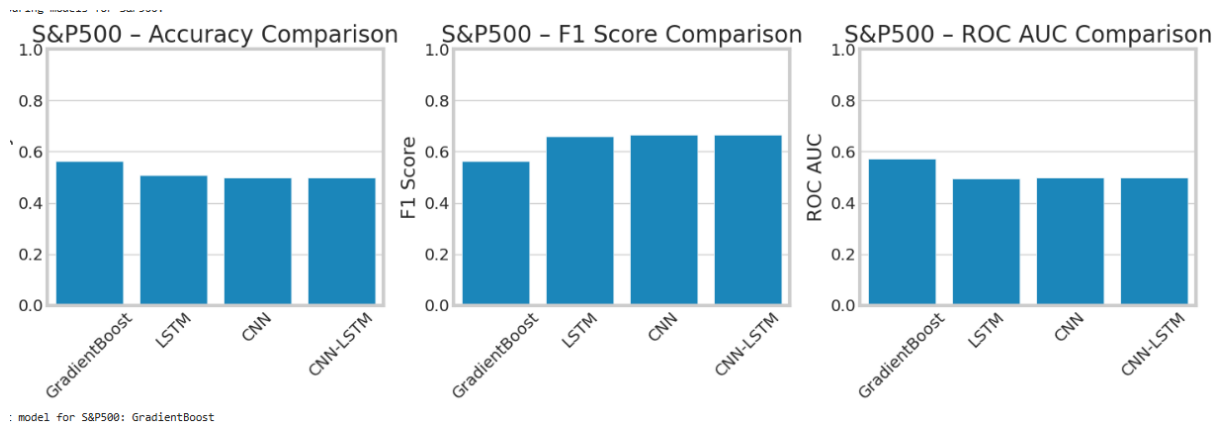


Figure 17 Results S&P 500

On similar patterns the results obtained from the Apple Stock data are also same.

- All the models showcased consistent results in terms of accuracy, but Gradient Boosting model attained a better result.
- All the deep learning models showcased a better F1-score for Apple.
- CNN-LSTM model attained a better AUC score of 0.5 compared to other models.

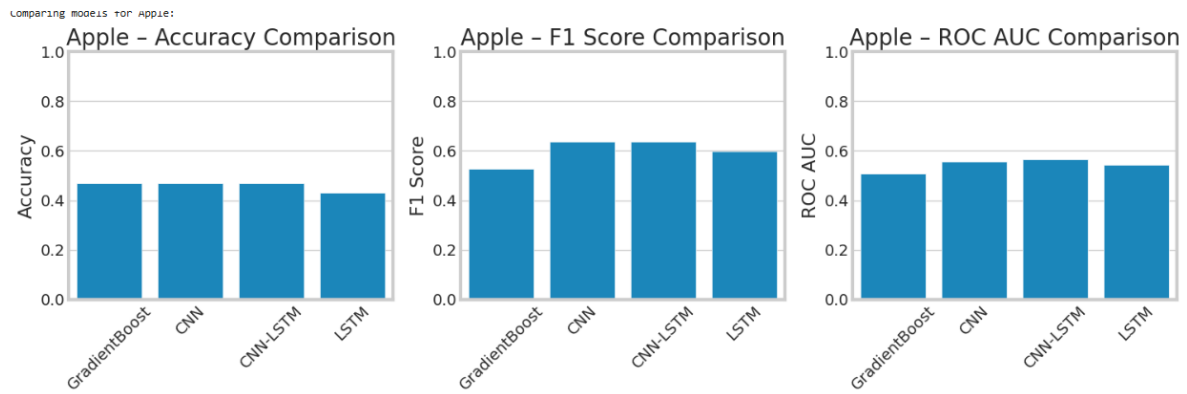


Figure 18 Results Apple

The last model evaluated here is on Microsoft and the results are similar to other stocks where all the other models showed consistent results and Deep learning-based models have better ROC AUC score and F1-score as compared to other models.

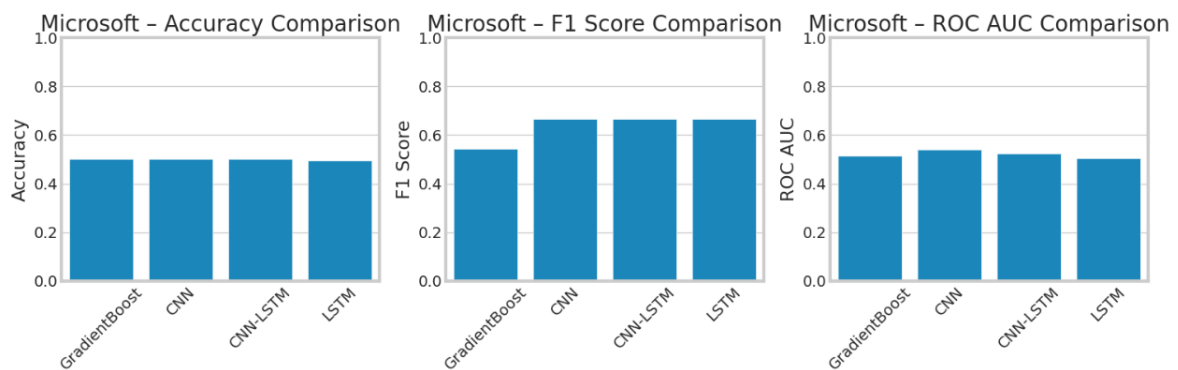


Figure 19 Results Microsoft

Hence the results showcase that the Gradient Boosting algorithm performed well for S&P 500 and for the individual stocks CNN+LSTM can be used. Gradient Boosting showcased better results because it is able to capture the market trends in a more generalized format. Now the models are explained based on XAI techniques.

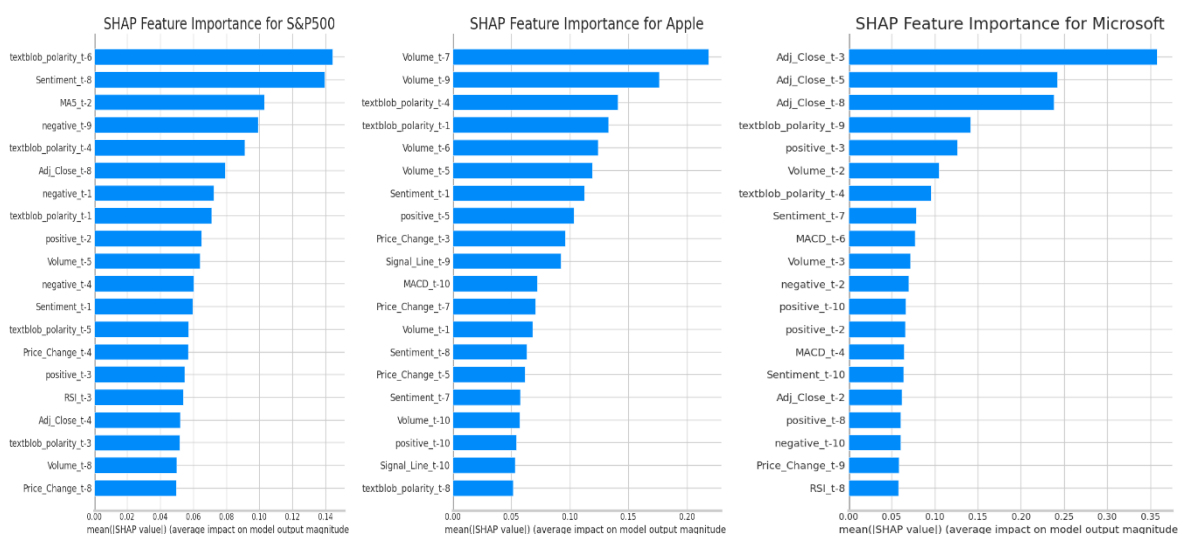




Figure 20 XAI

The models are explained based on the SHAP (Shapely Additive Explanations). This is a popular XAI technique based on the game theory (Villani et al. 2022; *SHAP for Time Series Event Detection | Towards Data Science* [no date]).

The output showcase the impact of features on the model output where the textblob polarity, sentiment score, moving average and ad close has high importance for S&P 500 stock but moderate to less moderate for the Apple and Microsoft. The volume of stock has very high impact for Apple stock and Price showcase high impact for the Microsoft stock. All these features contribute differently to the stock values based on the data available. Out of all the important features these models are also used for prediction, and it was observed that the models were able to predict the Up moment for S&P 500 but in the next sequence the outputs were incorrect.

Similarly, for Apple the True labels were correctly predicted and the same for Microsoft also. At the end the study concludes that the Gradient Boost algorithm performs well as compared to other deep learning models, showcasing the supremacy of machine learning models over deep learning models. the maximum accuracy in the dataset was not above 0.6 showcasing noise and non-linear nature of the data.

## CHAPTER 5– CONCLUSION AND FUTUREWORK

### 5.1 Conclusion

Traditional fundamental and technical analysis in stock price prediction have several limitations. Our research aimed to enhance the stock price forecasting by integrating financial data as well as sentiment analysis from news data using the CRISP-DM model. The study was driven by the need to combine text data with a time series analysis which was not possible in traditional models.

The data was collected from the Yahoo Finance API, Reddit News and processed differently to be used by different machine learning and deep learning models. Three different models are trained in the data set that are gradient boosting LSTM, LSTM+CNN. The financial news data is text data and is cleaned using the VADER and TextBlob to find the sentiment of the data which is then integrated with the time-series data. All the stock price doesn't show much correlation with any of the technical indicators and to increase the predictive power the news dataset was combined. All the three models showcased different results and were able to predict the next day movement either up or down. Gradient Boosting algorithm is the major baseline model which has performed well as compared to other algorithms and showcased adding the news sentiment improve the predictive power. All the research questions asked are answered as follows:

**RQ1:** Adding the news dataset tend to increase the accuracy of the model. Financial metrics when combined with the news dataset are able to enhance the market prediction. However, the results were satisfactory as data was non-linear and had noise. But the study showcased that adding the news sentiment data provide better context as compared to traditional models.

**RQ2:** Generally deep learning models showcase better results when combined with the text data and time series data, but here machine learning models showcased better results as compared to deep learning models.

**RQ3:** News sentiment tend to impact the stock market price Solely. As confirmed from the XAI techniques that the sentiment polarity adds more to the prediction of stock price when combined with Time Series data. Our research has found the sentiment using the VADER and TextBlob which can provide better predictions as compared to traditional models.

### 5.2 Future Work

The study showcased integrating the sentiment lead to predicting the stock price. But there are further research areas that can be explored in future.

- To provide better sentiment landscape data from Twitter, Reddit and earnings call transcript can be used.
- To predict better and help the analyst real-time dashboard can be built that moves as the stock price move daily.

- To understand the sentiment XAI techniques can be extended and find the sentiment features that influence output.
- Advance architectures like BERT and FinBERT can be used in the research to find sentiment analysis and forecast the results.
- Apply the model to different countries around the world to build a better and generalized model for more robustness.

Hence the study showcased a foundational step in reducing the gap of stock forecasting by adding sentiment scores to the Time Series data and validates the potential of these models to predict the stock price using news sentiment data.

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